25901837 Scheme for Valuation
MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-BVOC (HONOURS) Regular EXAMINATION April 2026
Second SEMESTER
Multi-Disciplinary Courses (MDC) - MG2MDCBFS100 – BUSINESS STATISTICS
(2025 ADMISSION ONWARDS)
Duration: 1 hour Maximum Marks: 35
Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.
Part A
VERY SHORT ANSWER
Answer any 5 questions. Each question carries 2 marks

1. How will you calculate Harmonic Mean in individual series? [U] / [CO3]

Harmonic Mean in an individual series is calculated by dividing the number of observations by the sum of reciprocals of the observations. The formula is:

$$HM = \frac{N}{\sum \frac{1}{X}}$$

It is mainly used for averaging rates and ratios.

2. State the formula for calculating Quartile Deviation in continuous series. [U] / [CO3]

Quartile Deviation measures the dispersion of the middle 50% of data. The formula is:
QD= (Q3 – Q1)/2
It is also known as Semi-Interquartile Range.

3. State the formula for calculating Karl Pearson's Coefficient of Correlation. [E] / [CO2]

Karl Pearson's Coefficient of Correlation measures the degree of linear relationship between two variables. The formula is:

$$r = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sqrt{\sum(X-\bar{X})^2 \sum(Y-\bar{Y})^2}}$$

The value of r lies between -1 and +1.

4. State Linear Regression. [E] / [CO2]

Linear regression is a statistical method used to study the relationship between two variables. It helps predict the value of one variable based on the value of another variable. The relationship is represented by a straight-line equation.

5. Define Statistics. [U] / [CO1]

Statistics is the science of collecting, organizing, presenting, analyzing, and interpreting numerical data. It helps in decision-making and drawing meaningful conclusions from data. Statistics is widely used in business, economics, and research.

6. State the situation of ill-defined mode. [U] / [CO3]

Mode becomes ill-defined when no value repeats frequently in a dataset or when two or more values occur with the same highest frequency. In such situations, it is difficult to identify a unique mode. The distribution may then be called bimodal or multimodal.

7. List out the different measures of central tendency (any two). [U] / [CO3]

Arithmetic Mean and Median are two important measures of central tendency. These measures help identify the central or average value of a dataset. Other measures include Mode and Geometric Mean.

Part B
SHORT ANSWER
Answer any 5 questions. Each question carries 5 marks

8. Calculate regression on Y on X [E] / [CO2]

x	y
5	4
6	5
7	8
3	2
2	1

Solution: $Y=-1.883+1.279X$

9.

X	f
5	2
10	4
15	6
20	5
25	3

Find the median. [An] / [CO4]

X	f	C.F.
5	2	2
10	4	6
15	6	12
20	5	17
25	3	20

The 10th item lies in the cumulative frequency 12, corresponding to X=15.
Median is 15

10. State the merits and demerits of Range. [U] / [CO4]
Range is the simplest measure of dispersion and is calculated as the difference between the highest and lowest values in a dataset. One major merit of range is that it is easy to calculate and understand. It helps in obtaining a quick idea about the variability of data. However, range is affected only by extreme values and ignores all other observations in the series. Therefore, it is considered less reliable for detailed statistical analysis.

11. How can statistics be misused intentionally? [U] / [CO1]
Statistics can be misused intentionally by presenting incomplete or misleading data to support false conclusions. People may manipulate graphs, averages, or percentages to create a wrong impression among viewers. Biased sampling and selective use of data are also common methods of statistical misuse. Such misuse can influence public opinion, business decisions, and policy making unfairly. Therefore, statistics should be used honestly and interpreted carefully.

12. There are three shifts in a factory. The mean wage of first shift is Rs107.75 and has 60 workers. In second shift and third shift there are 40 and 50 workers respectively. Their average wage is Rs 128.40 and Rs110. Find Combined Mean [A] / [CO4]

Shift	Number of Workers (f)	Mean Wage
First	60	107.75
Second	40	128.40
Third	50	110

Combined Mean Wage = **Rs. 114.01**

13. Define regression. State the important features. [E] / [CO2]

Regression is a statistical technique used to study the relationship between two or more variables and to predict the value of one variable based on another. It helps in measuring the effect of an independent variable on a dependent variable. Regression analysis is widely used in business, economics, and research for forecasting and decision-making. One important feature of regression is that it establishes a functional relationship between variables through regression equations. Another feature is that it helps in prediction and estimation, and regression coefficients indicate the direction and degree of relationship between variables.

14. Calculate Rank Correlation from the following ranks

1	4
5	8
4	7
8	6
9	5
6	9
10	10
7	3
3	2
2	1

[E] / [CO2]

X	Y	d = X−Y	(d^2)
1	4	-3	9
5	8	-3	9
4	7	-3	9
8	6	2	4
9	5	4	16
6	9	-3	9
10	10	0	0
7	3	4	16
3	2	1	1
2	1	1	1

Formula is $1 - \frac{6 \sum d^2}{n(n^2-1)}$

Rank Correlation Coefficient is: **0.552**